Database Management System – Question Bank

Module 1 & Module 2 | 20 Questions

Module 1 – Database Fundamentals, Architecture and ER/EER Modelling

Q1. Explain the limitations of a traditional file-processing system. Compare it with a DBMS with respect to **data redundancy, data inconsistency, data security, data sharing and program-data independence**. Give a suitable real-world example.

Q2. Draw and explain the **three-schema architecture of a DBMS**. Clearly show the **external, conceptual and internal levels** and explain how the architecture provides **logical and physical data independence**.
Draw a neat labelled diagram.

Q3. Consider an online shopping application where customers, products, orders and payments are managed through a database. Explain how the DBMS provides the following characteristics:

- Data abstraction
- Data sharing
- Data security
- Concurrency control
- Data integrity

Q4. Draw and explain the **1-tier, 2-tier and 3-tier DBMS architectures**. Compare them with respect to components, communication and suitability for modern web-based applications.
Use suitable architecture diagrams.

Q5. Identify and explain the different types of **database users**. Discuss the responsibilities of a **DBA, database designer, application programmer and end user** with a suitable organizational representation.

Q6. A hospital wants to develop an Electronic Health Record (EHR) system containing patient details, doctors, departments, appointments, prescriptions and medical reports. Explain the **role of a database system in healthcare** and identify at least five important data-management requirements of such a system.

Q7. Consider the following university scenario:

A university has several departments. Each department offers several courses. Students can enroll in multiple courses, and each course can have many students. Each student has a Student_ID, Name, Email and Phone. Each course has Course_ID, Course_Name and Credits.

Identify:

1. Entities
2. Attributes
3. Relationship sets
4. Primary keys
5. Cardinality constraints

Draw a suitable **ER diagram**.

Q8. Explain different types of attributes in an ER model with suitable examples:

- Simple attribute
- Composite attribute
- Single-valued attribute
- Multivalued attribute
- Derived attribute

Represent each type using appropriate **ER notation**.

Q9. Design an **ER diagram for a Hospital Management System** containing the following entities:

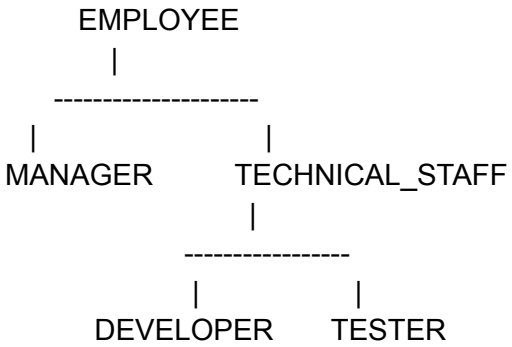
Patient, Doctor, Department, Appointment, Prescription and Medicine

The system should support:

- A patient can have multiple appointments.
- A doctor can attend multiple appointments.
- A doctor belongs to one department.
- An appointment may result in a prescription.
- A prescription may contain multiple medicines.

Clearly represent **cardinality and participation constraints**.

Q10. Explain **specialization, generalization and aggregation** in EER modelling with suitable examples. For the following hierarchy, draw an EER diagram:



Also explain how **disjointness and completeness constraints** can be applied to this hierarchy.

Module 2 – Relational Model, Relational Algebra and Normalization

Q11. Explain the following terms of the **relational model** with a suitable relation:

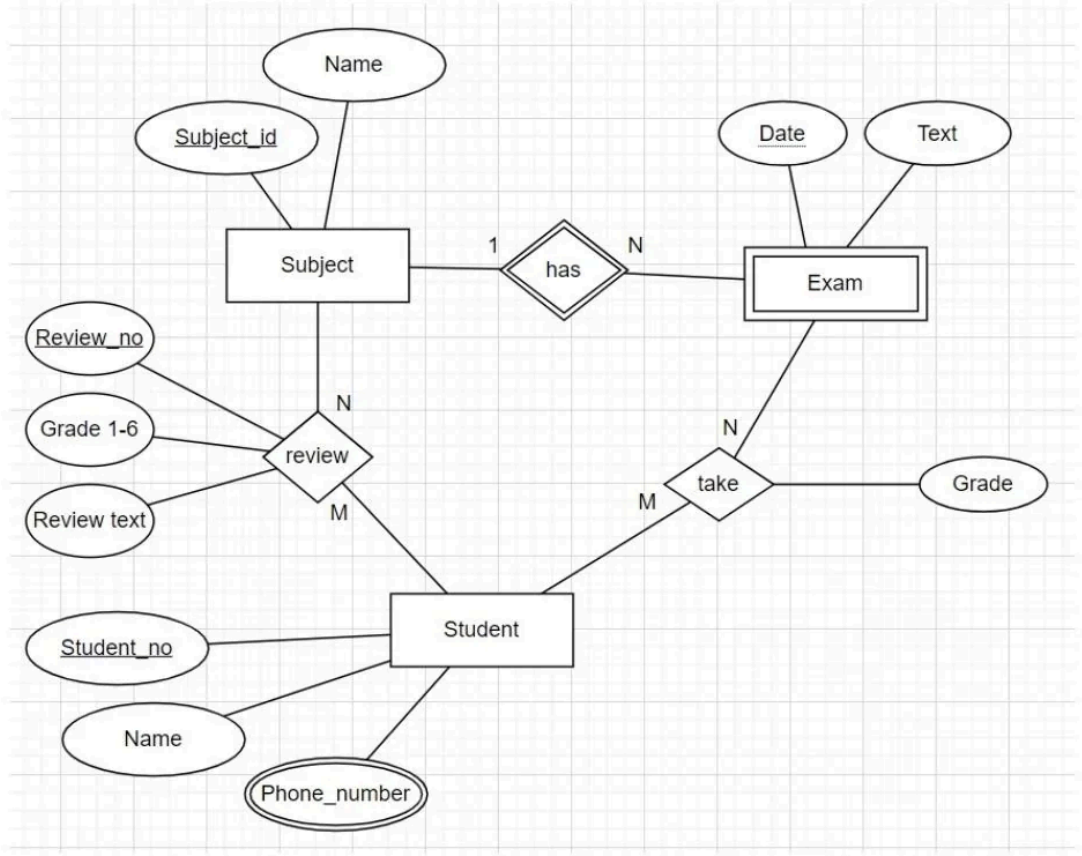
- Relation
- Tuple
- Attribute
- Domain
- Degree
- Cardinality

Consider:

Sid	Name	Branch	CGPA
S101	Amit	CE	8.5
S102	Neha	IT	9.1
S103	Ravi	CE	7.8

Determine the **degree and cardinality** of the relation.

Q12. Consider the following ER model description:



Convert this ER model into an appropriate **relational schema**. Clearly identify:

- Primary keys
- Foreign keys
- Relationship table

Q13. Explain the rules for converting an **ER/EER model into a relational schema**. Demonstrate the mapping using a suitable example containing:

- Strong entity
- Weak entity
- 1:N relationship
- M:N relationship

Q14. Consider the following relations:

STUDENT(Sid, Name, Branch, CGPA)
COURSE(Cid, Cname, Credits)
ENROLLMENT(Sid, Cid, Semester)

Write relational algebra expressions to find:

a) Students belonging to the CE branch.
b) Names of all students.
c) Students having CGPA greater than 8.5.
d) Students enrolled in the course `DBMS`.
e) Student names along with the courses in which they are enrolled.

Use appropriate operators such as **selection (σ), projection (π), Cartesian product (×), join and union**.

Q15. Explain the fundamental relational algebra operators **Selection, Projection, Union, Set Difference and Cartesian Product** with notation and suitable examples. Also explain the conditions required for **union compatibility**.

Q16. Consider the relations:

EMPLOYEE(Eid, Ename, Dept_ID, Salary)
DEPARTMENT(Dept_ID, Dept_Name, Location)

Write relational algebra expressions to:

a) Find employees working in the `Computer` department.
b) Find employees earning more than ₹60,000.
c) Display employee names and department names.
d) Find employees working in departments located in Mumbai.
e) Find departments having no employees.

Q17. Consider the following unnormalized relation:

STUDENT_COURSE
(
 Student_ID,
 Student_Name,
 Department,
 Course_ID,
 Course_Name,
 Instructor,
 Instructor_Email
)

Given the functional dependencies:

Student_ID → Student_Name, Department
Course_ID → Course_Name, Instructor

Instructor → Instructor_Email
(Student_ID, Course_ID) → all attributes

Identify the **candidate key** and normalize the relation up to **3NF**, showing every intermediate step.

Q18. Explain the need for normalization. Using a suitable example, demonstrate the following anomalies in an unnormalized relation:

- Insertion anomaly
- Update anomaly
- Deletion anomaly

Then explain how normalization eliminates these anomalies.

Q19. Consider the relation:

R(A, B, C, D, E)

with functional dependencies:

A → B
B → C
AC → D
D → E

Determine:

a) The closure of {A}
b) Candidate key(s) of R
c) Prime and non-prime attributes
d) Whether R is in 2NF
e) Whether R is in 3NF
f) Decompose R into appropriate relations if required.

Show all steps clearly.

Q20. Consider the relation:

R(Student_ID, Course_ID, Instructor, Room, Textbook)

with the following dependencies:

(Student_ID, Course_ID) → Instructor, Room

Instructor → Room

Course_ID → Textbook

a) Identify the candidate key.
b) Determine the highest normal form satisfied by the relation.
c) Decompose the relation into **BCNF** relations.
d) Check whether the decomposition is **lossless**.
e) Discuss whether the decomposition preserves the given functional dependencies.